Problem Set V, Macroeconomics III
University of Copenhagen

March 2026

Exercise 1 – The Frisch elasticity of labor supply

Consider a household that lives for two periods and values leisure. The household optimization problem is given by

$$\max_{c_0,c_1,h_0,h_1} \ln(c_0) - \frac{h_0^{1+\varphi}}{1+\varphi} + \beta\left[\ln(c_1) - \frac{h_1^{1+\varphi}}{1+\varphi}\right], \quad \varphi > 0, \varphi \neq 1$$
$$\text{s.t. } c_0 + \frac{c_1}{1+r} = w_0 h_0 + \frac{w_1 h_1}{1+r}$$

a) Find the first order conditions.

b) Combine the first order conditions to characterize the intertemporal choice of leisure, i.e., the slope of hours worked, h_1/h_0, as a function of the wage path, w_1/w_0. Show how higher values of φ lead the household to choose a smoother leisure path over time. Interpret.

Exercise 2 – Labor supply and the effects of wealth and taxes

Consider a consumer with utility function defined over consumption, c and leisure, l: $u(c,h) = \log(c) - B\frac{(h)^{1+\nu}}{1+\nu}$. The consumer's budget constraint reads

$$c = wh(1-\tau) + a + T$$

where $1 - l = h$ is hours worked (there is a total time endowment of 1 unit, as in the lecture slides), w is the wage, a is an amount of wealth, τ is a proportional tax rate, and T is a transfer from the government.

a) Consider $a = \tau = T = 0$. State and solve the consumer's maximization problem. How does the wage affect hours worked?

b) Suppose a is nonzero (while maintaining taxes and transfers at zero). How is the consumer's labor choice affected by a

c) Suppose $a = 0$ again, and that the government's revenue from taxing labor income is thrown in the Baltic Sea. How does the level of τ affect hours worked?

d) Suppose $a = 0$, that $\tau > 0$, and that the government's revenue from taxing labor income is handed back to the consumer (in the form of something that can be compared to general consumption resources), so that $T = w\tau h$, where h is the consumer's equilibrium choice of hours. The transfer, however, is lump-sum: by choosing different levels of h, the consumer cannot influence the amount of transfer (this is justified by interpreting the economy as being inhabited by many consumers of the same type). Solve for labor-supply as a function of τ.

Exercise 3 – Working past retirement

Consider an economy with overlapping generations. At each point in time, two generations are alive at the same time, young and old. Each individual lives for two periods, and their lifetime utility is given by

$$U = \log(c_{y,t}) + \beta \log(c_{o,t+1})$$

where $c_{y,t}$ is consumption when young, in period t, and $c_{o,t+1}$ is consumption when old. The parameter $0 < \beta < 1$ describes their patience. Labor is supplied inelastically and hence does not enter the utility function.

An individual's budget constraints are given by

$$\begin{aligned} c_{y,t} + s_t &= w_t \\ c_{o,t+1} &= \Gamma w_{t+1} + (1 + r_{t+1}) s_t \end{aligned}$$

where w_t is the prevailing wage in period t, r_{t+1} is the interest rate paid on savings between periods t and $t+1$, and s_t is the amount saved. Finally, $\Gamma \in \{0, 1\}$ is an indicator variable that takes the value of 1 if the government requires workers to work when old, and zero if workers may only work in the first period of their lives.

a) Set up the Lagrangian for this problem and solve for the first order conditions.

b) Show that the consumer's lifetime consumption path is described by an Euler equation. Explain, using economic intuition, how a change in the interest rate affects the

path of consumption over time.

c) Combining the Euler equation with the budget constraints, show that optimal savings, as a function of wage rates, the interest rate and patience, can be expressed as

$$s_t = \frac{\beta(1+r_{t+1})w_t - \Gamma w_{t+1}}{(1+\beta)(1+r_{t+1})} \tag{1}$$

When $\Gamma = 0$, savings are independent of the interest rate. Explain why.

Workers in the model save into capital K_t, which is used for production and does not depreciate. Perfectly competitive firms use it to produce according to a Cobb-Douglas production function, combining it with aggregate labor L_t. They maximize profits:

$$\Pi = K_t^{\alpha} L_t^{1-\alpha} - r_t K_t - w_t L_t \quad 0 < \alpha < 1$$

where r_t represents the interest rate paid on capital and w_t represents wages paid to all workers. There is no population growth, and the total mass of people in the economy who are young is $P_y = 0.5$. The total mass of individuals is normalized such that $P = P_y + P_o = 1$.

d) Show, by maximizing the firm's profits, that wages and interest rates satisfy

$$w_t = (1-\alpha)k_t^{\alpha}(1+\Gamma)^{-\alpha}$$
$$r_t = (\alpha)k_t^{\alpha-1}(1+\Gamma)^{1-\alpha}$$

where $k_t = K_t/P_y$ indicates **capital-per-young**. *(Hint: remember that the size of the workforce L_t depends on Γ, i.e., on whether the government forces the old to work.)*

e) Impose asset market clearing ($s_t = k_{t+1}$) and find the steady state for capital when $\Gamma = 0$, as a function of β and α. How does the steady state level of capital depend on α? Give an economic intuition.